Activity No. 13	
Working with Files and Directories in Linux	
Course Code: CPE 201A	**Program:CpE**
Course Title: COMPUTER SYSTEM ADMINISTRATION AND TROUBLESHOOTING	**Date Performed:10/22/26**
Section: CPE11S4	**Date Submitted:10/22/26**
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1. Objective/s:

This activity aims to navigate and manage files and directories using command line interface.

2. Intended Learning Outcome/s:

The students should be able to:

2.1 Demonstrate how to navigate home and system directories
2.2 Demonstrate how to how to list files and directories.
2.3 Demonstrate how to use globbing to manipulate files and directories.
2.4 Demonstrate how to create, move and delete files and directories.

3. Discussion:

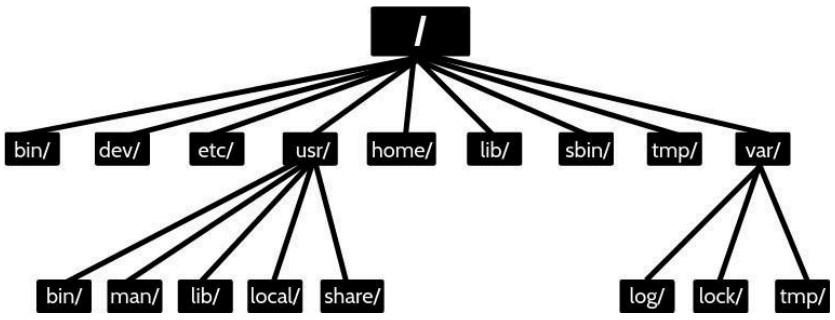
Files and Directories

The Linux filesystem is similar to other operating system's filesystems in that it contains *files* and *directories*. Files contain data such as human-readable text, executable programs, or binary data that is used by the computer.

Directories are used to create organization within the filesystem. Directories can contain files and other directories.

Directory Structure

On a Windows system, the *top level* of the directory structure is called ***My Computer.*** The Linux directory structure, called a ***filesystem***, also has a top level called the ***root directory*** (symbolized by the **slash /character**).



Home Directory

On most Linux distributions there is a directory called **home** under the root /directory. Under this **/home** directory there is a directory for each user on the system. When a user opens a shell, they should

automatically be placed in their home directory. The user has the full control to create and delete additional files and directories in their home directory. Most other directories in a Linux filesystem are protected with *file permissions*. The home directory has a special symbol used to represent it, the **tilde ~ character**.

The directory name is the same as the name of the user. So, a user named sysadmin would have a *home directory* called /home/sysadmin:

File and Directory Names

File and directory names in Linux can contain lower case and upper case letters, numbers, spaces and special characters. However, since many special characters have a special meaning in the Linux shell, it is good practice to not use spaces or special characters when naming files or directories. Spaces, for example, need the *escape character* \ to be entered correctly

Paths

A path is a list of directories separated by the / character. There are two types of paths: *absolute* and *relative*. For example, /home/sysadmin is a path to the home directory

A. Absolute Paths

Absolute paths allow the user to specify the exact location of a directory. Absolute paths always starts at the root directory, and therefore it always begins with the / character.

B. Relative Paths

A relative path gives directions to a file relative to the current location in the filesystem. The user must currently be in a directory that contains objects in the path. Relative paths start with the name of a directory.

Special Relative Paths

The single period . character always represents the current directory.
Two period .. characters always represents one directory higher relative to the current directory, sometimes referred to as the parent directory.

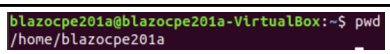
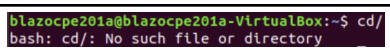
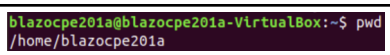
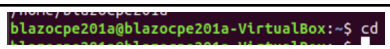
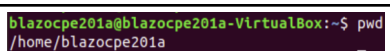
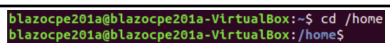
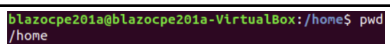
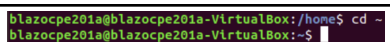
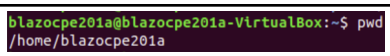
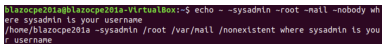
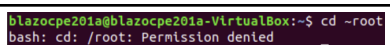
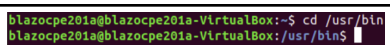
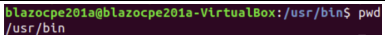
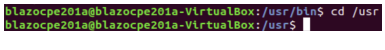
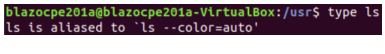
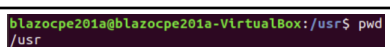
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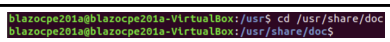
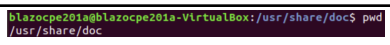
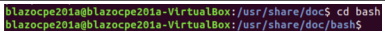
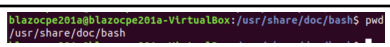
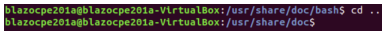
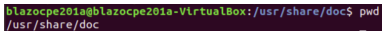
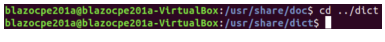
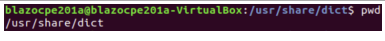
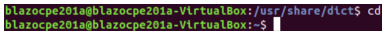
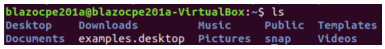
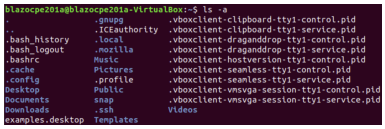
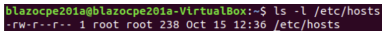
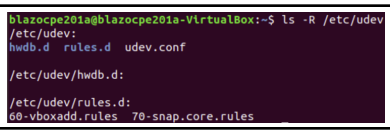
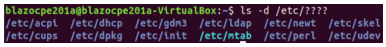
Personal Computer with installed Virtual Box
Ubuntu Server or Desktop virtual machine

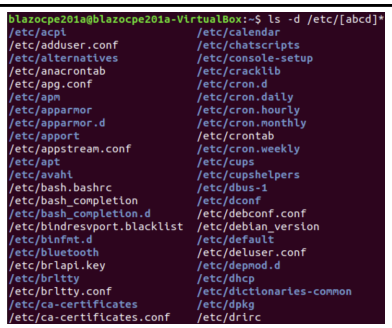
5. Procedure:

1. Login using your username and password.
2. Use terminal emulator application (if you are using desktop version)
3. Execute the following commands to navigate files and directories. Copy a screenshot as output after you execute the given command. Create a brief explanation of the command.

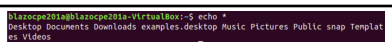
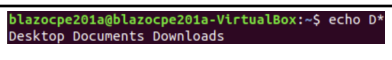
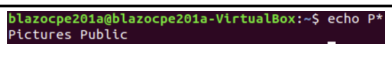
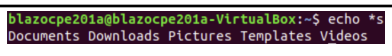
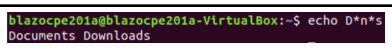
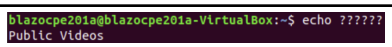
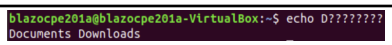
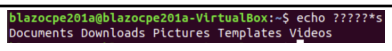
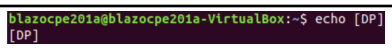
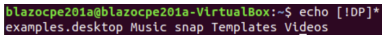
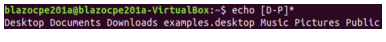
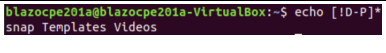
Command	Screenshot	Explanation

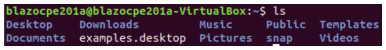
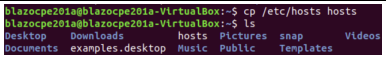
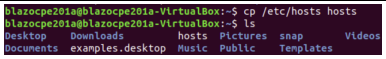
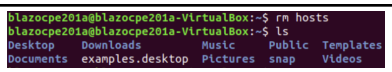
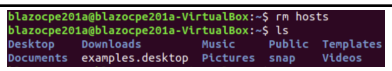
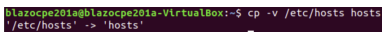
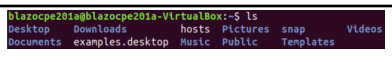
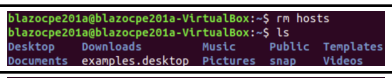
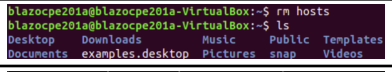
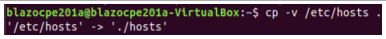
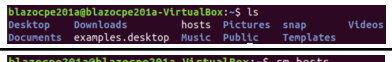
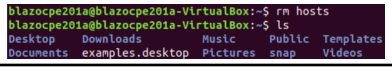
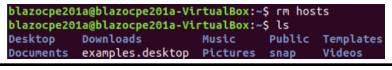
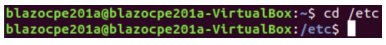
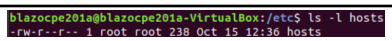
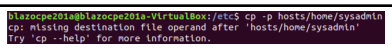
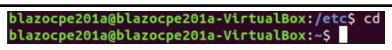
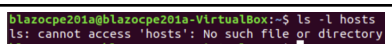
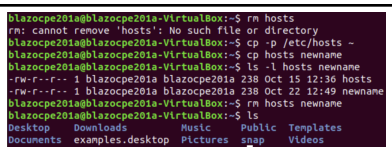
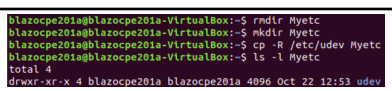
1. pwd		Displays the absolute path of the current working directory (/home/blazocpe201a).
2. cd /		Attempts to change directory to / (root), but throws an error due to missing space after 'cd'.
3. pwd		Confirms the current working directory remains /home/blazocpe201a.
4. cd		Changes the directory to the user's home directory by default.
5. pwd		Confirms current location is the user's home directory (/home/blazocpe201a).
6. cd /home		Navigates to the /home directory.
7. pwd		Prints the current directory path, showing /home.
8. cd ~		Navigates back to the current user's home directory using the tilde (~) symbol.
9. pwd		Verifies the current directory is /home/blazocpe201a.
10. echo ~ ~sysadmin ~root ~mail ~nobody where sysadmin is your username		Prints tilde expansions for the current user and specified system account directories.
11. cd ~root		Attempts to enter root's home directory (~root), resulting in 'Permission denied'.
12. cd /usr/bin		Changes working directory to /usr/bin.
13. pwd		Confirms current working directory is /usr/bin.
14. cd /usr		Changes working directory to /usr.
15. type ls		Displays information about the 'ls' command, showing it is aliased to 'ls --color=auto'.
16. pwd		Confirms current location is /usr.

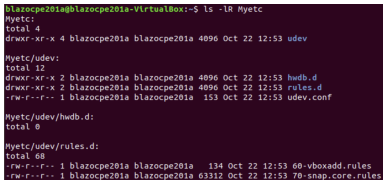
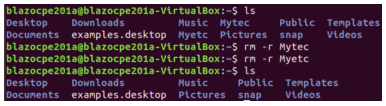
17. cd /usr/share/doc		Changes current directory to /usr/share/doc.
18. pwd		Confirms current location is /usr/share/doc.
19. cd bash		Enters the 'bash' directory located inside /usr/share/doc.
20. pwd		Confirms current location is /usr/share/doc/bash.
21. cd ..		Navigates up one level to the parent directory (/usr/share/doc).
22. pwd		Confirms current location is /usr/share/doc.
23. cd ../dict		Navigates to the parent directory's sibling folder 'dict' (/usr/share/dict).
24. pwd		Confirms current location is /usr/share/dict.
25. cd		Returns to the user's home directory.
26. ls		Lists visible files and subdirectories in the current directory.
27. ls -a		Lists all files and directories, including hidden ones starting with a dot (.).
28. ls -l /etc/hosts		Displays detailed file permissions, owner, size, and modification date of /etc/hosts.
29. ls -R /etc/udev		Recursively lists all files and subdirectories under /etc/udev.
30. ls -d /etc/s*		Lists directories in /etc that begin with the letter 's'.
31. ls -d /etc/????		Lists entries in /etc that have exactly four characters in their name.

32. ls –d /etc/[abcd]*		Lists directories in /etc starting with 'a', 'b', 'c', or 'd'.

4. Execute the following commands to manage files and directories. Copy a screenshot as output after you execute the given command. Create a brief explanation of the command.

Command	Screenshot	Explanation
1. echo *		Uses wildcard '*' to echo all visible files and directories in the current folder.
2. echo D*		Echoes names of files or directories starting with capital 'D'.
3. echo P*		Echoes names of files or directories starting with capital 'P'.
4. echo *s		Echoes names of files or directories ending with 's'.
5. echo D*n*s		Echoes names matching the pattern: start with 'D', contain 'n', and end with 's'.
6. echo ??????		Echoes items with exactly six characters in their name.
7. echo D????????		Echoes items starting with 'D' followed by exactly eight characters.
8. echo ?????*s		Echoes items starting with at least 5 characters and ending with 's'.
9. echo [DP]*		Echoes files/directories starting with either 'D' or 'P'.
10. echo [!DP]*		Echoes files/directories that do NOT start with 'D' or 'P'.
11. echo [D-P]*		Echoes items starting with any letter in the range D through P.
12. echo [!D-P]*		Echoes items starting with any letter outside the range D through P.

13. ls		Lists contents of current working directory.
14. cp /etc/hosts hosts		Copies /etc/hosts to a local file named 'hosts' in current directory.
15. ls		Verifies that the file 'hosts' was created in the current directory.
16. rm hosts		Removes the local file 'hosts'.
17. ls		Confirms deletion of 'hosts'.
18. cp –v /etc/hosts hosts		Copies /etc/hosts with verbose output (-v) indicating the action taken.
19. ls		Lists directory contents showing 'hosts'.
20. rm hosts		Removes 'hosts' file.
21. ls		Confirms 'hosts' file removal.
22. cp –v /etc/hosts .		Copies /etc/hosts to current directory (.) with verbose output.
23. ls		Confirms presence of 'hosts'.
24. rm hosts		Removes 'hosts' file.
25. ls		Confirms removal of 'hosts'.
26. cd /etc		Changes working directory to /etc.
27. ls -l hosts		Displays detailed listing of hosts file in /etc.
28. cp –p hosts/home/sysadmin		Fails due to missing space between source and destination (hosts/home/sysadmin).
29. cd		Changes directory back to home.
30. ls –l hosts		Fails to locate 'hosts' in home directory because previous copy command failed.
31. rm hosts 32. cp -p /etc/hosts ~ 33. cp hosts newname 34. ls –l hosts newname 35. rm hosts newname		Demonstrates attempting deletion, preserving file attributes during copy (-p), duplicating files, listing details, and cleaning up.
36. mkdir Myetc 37. cp –R /etc/udev Myetc		Creates 'Myetc' directory, copies /etc/udev recursively (-R) into it,

38. ls –l Myetc 39. ls –lR Myetc		and lists the directory recursively.
40. ls 41. rm -r Myetc 42. ls		Lists contents, recursively deletes 'Myetc' directory and contents (-r), and verifies deletion.
43. touch premove 44. ls 45. mv premove postmove 46. ls 47. rm postmove		Creates 'premove' using touch, renames/moves it to 'postmove' using mv, and deletes it with rm.

6. Supplementary Activity:

Copy screen shot(s) of the following tasks:

1. Create a directory CPE201A in the Documents directory.

```
blazocpe201a@blazocpe201a-VirtualBox:~$ cd ~/Documents
blazocpe201a@blazocpe201a-VirtualBox:~/Documents$ mkdir CPE201A
blazocpe201a@blazocpe201a-VirtualBox:~/Documents$ ls
CPE201A
```

2. Create a directory <lastname> and <firstname> under CPE201A directory.

```
blazocpe201a@blazocpe201a-VirtualBox:~/Documents$ cd ~/Documents/CPE201A
blazocpe201a@blazocpe201a-VirtualBox:~/Documents/CPE201A$ mkdir Blazo Joaquin
```

3. Create two text files: (1) A.txt and (2) B.txt in the <lastname> directory.

```
blazocpe201a@blazocpe201a-VirtualBox:~/Documents/CPE201A$ cd ~/Documents/CPE201
A/Blazo
blazocpe201a@blazocpe201a-VirtualBox:~/Documents/CPE201A/Blazo$ touch A.txt B.t
xt
```

4. Copy the content of <lastname> directory to <firstname> directory.

```
blazocpe201a@blazocpe201a-VirtualBox:~/Documents/CPE201A/Blazo$ cp -r ~/Documen
ts/CPE201A/Blazo/. ~/Documents/CPE201A/Joaquin/
```

5. Delete the content of <lastname> directory.

```
blazocpe201a@blazocpe201a-VirtualBox:~/Documents/CPE201A/Blazo$ rm -r ~/Documen
ts/CPE201A/Blazo/*
```

6. Create a directory CPE201A_backup in the Documents directory.

```
blazocpe201a@blazocpe201a-VirtualBox:~/Documents/CPE201A/Blazo$ mkdir ~/Documen
ts/CPE201A_backup
```

7. Copy the content of CPE231 to CPE201A_backup directory.

```
blazocpe201a@blazocpe201a-VirtualBox:~/Documents/CPE201A/Blazo$ cp -r ~/Documen
ts/CPE201A/. ~/Documents/CPE201A_backup/
```

8. Remove the CPE201A directory.

```
blazocpe201a@blazocpe201a-VirtualBox:~/Documents/CPE201A/Blazo$ rm -r ~/Documen
ts/CPE201A
```

7. Conclusion:

This activity helped me learn how to navigate and manage files and directories in Linux using commands. I learned how to use paths, search for files, and create, copy, rename, and delete files and directories. These skills are useful in Linux system administration.

8. Assessment (Rubric for Laboratory Performance):